

MTH101 QUIZ(1)

Lecture: 14 to 18

RIZ MUGHAL
SQA ENGINEER:

I'm providing 100% correct quiz solution.

You can visit my YouTube channel for more quiz solution, also final year project including project assignments, and viva.

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MTH101:Quiz 1

Question # 1 of 10 (Start time: 12:44:40 PM, 04 June 2021)

If f(x)=8x ,then f ' (2) is.....

Select the correct option

○	2
○	3
○	4
●	8

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MTH101:Quiz 1 Quiz Start Time

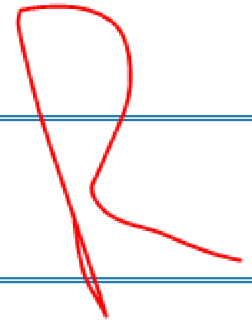
Question # 2 of 10 (Start time: 12:45:46 PM, 04 June 2021)

If a car travels 150 miles over a straight road in 3 hours, then its average velocity during the trip is .....

Select the correct option

- (●) 50m/h
- ( ) 30m/h
- ( ) 60m/h
- ( ) 40m/h

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Question # 3 of 10 (Start time: 12:46:26 PM, 04 June 2021)

If y=2x then instantaneous rate of change of 'y' w.r.t 'x' at 'x =-2' is .....

Select the correct option

○	3
○	-1
●	2
○	1

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Question # 4 of 10 (Start time: 12:46:56 PM, 04 June 2021)

d(-2x+3cosx)/dx=...

Select the correct option

- 2-3sinx
- -2-3sinx
- 2+3sinx
- -2+3sinx

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MTH101:Quiz 1 Quiz Start Time: 12:44 P

Question # 5 of 10 (Start time: 12:47:19 PM, 04 June 2021)

$If\ f(x) = (\sec x + x^4)^5$, *then the derivative of* $f(x)$ *is* _____.

Select the correct option

Reload

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- (●) $5(\sec x + x^4)^4(\sec(x)\tan(x) + 4x^3).$
- ( ) $5(\sec x + x^4)^4.$
- ( ) $5(\sec x + x^4)^4(\sec(x)\tan(x) - 4x^3).$
- ( ) None of these.

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Question # 6 of 10 (Start time: 12:47:41 PM, 04 June 2021)

For the curve f(t) = -23t+25, the slope of the tangent line at point 't = -1' is _____.

Select the correct option

- -25
- 23
- 25
- -23

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Question # 7 of 10 (Start time: 12:48:01 PM, 04 June 2021)

The derivative of sin(2x) is

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Select the correct option

- ○ Cosx
- ○ Cos2x
- ○ 2Cos(x)
- ● Cos(2x).2

MTH101:Quiz 1 Quiz Start Time: 12:44 PM, 04 Jun

Question # 8 of 10 (Start time: 12:48:22 PM, 04 June 2021) Total M

$If\ f(x) = secx\ and\ g(x) = csc(x),\ then\ the\ derivative\ of\ f(x) \times g(x)\ is$

Select the correct option

Reload Math Equa

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- None of these.
- $-\sec x \csc x \cot x + \csc x \sec x \tan x$
- $-\sec x \csc x \cot x - \csc x \sec x \tan x$
- $\sec x \csc x \cot x + \csc x \sec x \tan x$

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Question # 9 of 10 (Start time: 12:48:50 PM, 04 June 2021)

The cotangent function is defined by Cot x= cosx/sinx

Select the correct option

- (●) True
- ( ) False

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MTH101:Quiz 1 Quiz Start Time: 12:44 P

Question # 10 of 10 (Start time: 12:49:34 PM. 04 June 2021)

Derivative of $\cos 4x$ *by* using *chain rule is* . . . . . .

Select the correct option

Reload

- ● $-4 \sin 4x$
- ○ $\sin 4x$
- ○ $-\sin 4x$
- ○ $4 \sin 4x$

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2nd account

MTH101:Quiz 1 Quiz Start Time: 03:57 PM, 04 Ju

Question # 1 of 10 (Start time: 03:57:12 PM, 04 June 2021) Total

If y=f(x) then instantaneous rate of change of y with respect to x at the point x0 is given by slope of tangent line at (x0,f(x0))

Select the correct option

- (●) True
- ( ) False

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MTH101:Quiz 1

Quiz Start Time: 03:5

Question # 2 of 10 (Start time: 03:57:53 PM, 04 June 2021)

$$If\ f(x) = \sqrt{x}\ and\ g(x) = 4\sqrt{x}, then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$$

Select the correct option

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●	None of these
○	2
○	4
○	x + 2

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MTH101:Quiz 1 Quiz Start Time: 03:57 PM

Question # 3 of 10 (Start time: 03:58:15 PM. 04 June 2021)

What is the derivative $\tan(x^2 + x^3)$?

Select the correct option

Reload Ma

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- $(2x^3 + 3x)\cos ec^2(x^2 + x^3)$
- None of these
- $(2x^3 + 3x)\sec^2(x^2 + x^3)$
- $(3x^2 + 2x)\sec^2(x^2 + x^3)$

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Quiz Start Time: 03:57 PM

Question # 4 of 10 (Start time: 03:58:35 PM, 04 June 2021)

The derivative of $(\sin(\cos x))$ *is* .....

Select the correct option

Reload Ma

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- $\cos(\cos x)$
- $\cos(\sin x)$
- $-\sin(\cos x).\sin x$
- $-\cos(\cos x).\sin x$

MTH101:Quiz 1 Quiz Start Time: 03:57 PM

Question # 5 of 10 (Start time: 03:58:54 PM, 04 June 2021)

$$If\ y(x) = \tan x,\ then\ y''(x)\ is$$

Select the correct option

Reload Ma

- ● $2sec^2(x)tan(x)$
- ○ $-6sec^2(x)tan(x)$
- ○ $-2sec^2(x)tan(x)$
- ○ $\sec x \tan x$

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MTH101:Quiz 1 Quiz Start Time: 03:

Question # 6 of 10 (Start time: 03:59:14 PM, 04 June 2021)

What is the derivative of $2x^2 \sin(x^5)$?

Select the correct option

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○	$-10x^2 \cos(x^5)$
●	None of these
○	$2x^2 \cos(x^5)$
○	$10x^2 \cos(x^5)$

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Quiz Start Time: 03:5

Question # 7 of 10 (Start time: 03:59:45 PM, 04 June 2021)

What is the derivative of 3cot (2x)

Select the correct option

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- $-3cosec^2(2x)$
- $-6cosec^2(2x)$
- None of these
- $6cosec^2(2x)$

Click to Save Answer & Move to

MTH101:Quiz 1

Question # 8 of 10 (Start time: 04:00:04 PM, 04 June 2021)

If f(x)=8x ,then f ' (2) is.....

Select the correct option

- (●) 8
- ( ) 3
- ( ) 2
- ( ) 4

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Question # 9 of 10 (Start time: 04:00:52 PM, 04 June 2021)

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- (●) Its average rate of change is 7.
- ( ) Its average rate of change is 9.
- ( ) None of these.
- ( ) Its average rate of change is 8.

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Question # 10 of 10 (Start time: 04:01:10 PM, 04 June 2021)

If $u(t) = 4\cos t$, *then* $u'(\frac{\pi}{2}) = \ldots$.

Select the correct option

- 2
- -4
- 4
- 9

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3rd account

MTH101:Quiz 1 Quiz Start Tim

Question # 1 of 10 (Start time: 04:10:09 PM, 04 June 2021)

For the curve: f(t) = -2t+3, the instantaneous rate of change of 'f(t)' at 't = 1' is _____.

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Select the correct option

- (●) -2
- ( ) 1
- ( ) -1
- ( ) 2

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Question # 2 of 10 (Start time: 04:10:37 PM, 04 June 2021)

A function .....be differentiable at the point of discontinuity.

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Select the correct option

- can
- cannot

MTH101:Quiz 1 Quiz Start Time: 04:10 PM.

Question # 3 of 10 (Start time: 04:10:56 PM, 04 June 2021)

$If\ f(x) = x \sin x^2, then\ which\ of\ the\ following\ is\ true\ about\ it$

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Select the correct option

Reload Mat

- ○ $Its\ derivative\ w.r.t\ x\ is\ x \cos x^2 + \sin x^2$
- ○ $Its\ derivative\ w.r.t\ x\ is\ 2x \cos x^2 + \sin x^2$
- ○ $None\ of\ these$
- ● $Its\ derivative\ w.r.t\ x\ is\ 2x^2 \cos x^2 + \sin x^2$

MTH101:Quiz 1

Quiz Start Time: 04:10 PM, 04 J

Question # 4 of 10 (Start time: 04:11:12 PM, 04 June 2021)

Total

$If\ f(x) = ax + bx + c,\ then\ its\ derivative\ with\ respect\ to\ x\ is\ \ldots\ldots\ldots\ldots$

Select the correct option

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- ○ $b + c$
- ● $a + b$

MTH101:Quiz 1 Quiz Start Time: 04

Question # 5 of 10 (Start time: 04:11:30 PM, 04 June 2021)

The derivative of function:
f(x) = -2(x-1) + {5/(x-1)} does not exist at
x = --------

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Select the correct option

○	-1
○	0
○	-5
●	1

MTH101:Quiz 1 Quiz Start Time: 04:10 PM, 04 June 2

Question # 6 of 10 (Start time: 04:11:47 PM, 04 June 2021) Total Mar

$$If\ f(x) = sinx\ and\ g(x) = x^2,\ then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$$

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Select the correct option

Reload Math Equatic

- $[x^2cos(x) + 2xsin(x)]/x^4$
- $[x^2cos(x) - 2xsin(x)]/x^2$
- None of these.
- $[x^2cos(x) - 2xsin(x)]/x^4$

MTH101:Quiz 1 Quiz Start Time: 04:10

Question # 7 of 10 (Start time: 04:12:05 PM, 04 June 2021)

What is the derivative of sin(x) at $x = \pi/2$

Select the correct option

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- (●) 0
- ( ) None of these
- ( ) 1
- ( ) 0.707

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MTH101:Quiz 1 Quiz Start Time: 04:10 PM,

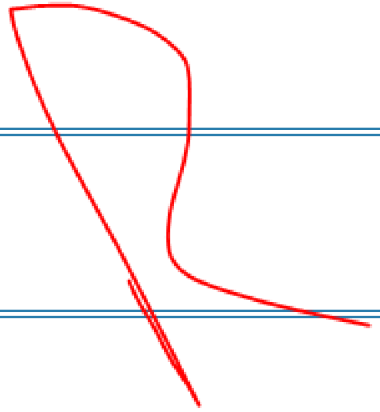
Question # 8 of 10 (Start time: 04:12:25 PM, 04 June 2021)

If (0,0) and (2,2) are any two pints on a curve then the slope of secant line through these points is

Select the correct option

- -1
- -2
- 2
- 1 (●)

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MTH101:Quiz 1 Quiz Start Time: 04:10

Question # 9 of 10 (Start time: 04:12:43 PM, 04 June 2021)

The function f(x) = |x| has no derivative at x = 0 because there is a _____ at that point.

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Select the correct option

- None of these
- Vertical tangent
- Corner
- Discontinuity

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Question # 10 of 10 (Start time: 04:13:01 PM. 04 June 2021)

Slope of secant line joining points (2,3) and (3,2) is____.

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Select the correct option

- -1
- None of these
- 1
- 2

Thank you for watching ☺

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